

Aerodynamic & Performance Analysis — Single-Engine Gasoline UAV

Prepared: 2026-06-14 · **Configuration:** high-aspect rectangular wing, twin-boom-style tubular fuselage, single tractor (puller) gasoline engine (DLE-120), 28×10 propeller.

Read this first. This is a first-order analytical estimate built from the supplied data sheet using standard handbook methods (component drag build-up, lifting-line induced drag, ISA atmosphere, energy-based fuel model). It is intended for design-trade and sanity-check purposes — **not** a substitute for wind-tunnel data, CFD, or flight test. All assumptions are stated explicitly so any number can be re-derived or corrected. Where the supplied data sheet is internally inconsistent, this is flagged.

1. Input parameters (as supplied)

Parameter	Value	Parameter	Value
Wingspan b	4.0 m	MTOW	50 kg
Wing area S	2.0 m² (4.0 × 0.5 m, rectangular)	Empty weight	19 kg
Root = tip chord	0.50 m	Max payload	20 kg
Airfoil	"NACA 25112" (see §2.3)	Fuel capacity	15 L
Engine	DLE-120 (120 cm³ gasoline twin)	Cruise burn	60 mL/min = 3.6 L/h
Propeller	28×10 tractor	V_min	81 km/h (22.5 m/s)
Static thrust	26 kgf (≈255 N)	V_nom	150 km/h (41.7 m/s)
Fuselage pod	Ø160 mm × 1.8 m	V_ne	180 km/h (50.0 m/s)
Tail boom	Ø100 mm × 1.0 m	Load factor	+4.8 / −2.4 g
Max endurance	10 h	Ferry range	600 km
Service altitude	500 m / 4000 m AMSL	Max altitude	5000 m

1.1 Derived geometry & loading

Quantity	Value	Note
Aspect ratio AR = b²/S	8.0	moderate-high; good for efficiency
Mean chord	0.50 m	constant (rectangular)
Wing loading W/S @ MTOW	245 N/m² = 25.0 kg/m²	high for this class → fast, but high stall speed
Power loading @ ~12 hp	4.17 kg/hp	well powered
Static thrust-to-weight	0.52	very strong for a fixed-wing
Fuel mass (gasoline ρ≈0.72)	10.8 kg	19 + 10.8 + 20 ≈ 49.8 kg ✓ closes to MTOW

The mass budget closes almost exactly: **19 kg empty + ~10.8 kg fuel + 20 kg payload ≈ 50 kg MTOW**. The airframe is therefore designed to 50 kg, not arbitrarily limited there.

2. Method & assumptions

2.1 Atmosphere (ISA)

Altitude	Temp	Density ρ	μ (Pa·s)	Speed of sound
0 m (ref)	15.0 °C	1.2250 kg/m ³	1.789e-5	340.3 m/s
500 m	11.8 °C	1.1673 kg/m³	1.774e-5	338.4 m/s
4000 m	−11.0 °C	0.8191 kg/m³	1.661e-5	324.6 m/s
5000 m	−17.5 °C	0.7361 kg/m ³	1.628e-5	320.5 m/s

Density ratio $\sigma(4000\text{ m})/\sigma(\text{SL}) = 0.669$ — this is the single biggest driver of the differences between the two altitude cases.

2.2 Drag model

Parasite drag from a **component build-up** (Hoerner / Raymer form factors, turbulent flat-plate skin friction $C_f = 0.455/(\log_{10}Re)^{2.58}$). Induced drag from $C_{Di} = k \cdot C_L^2$ with $k = 1/(\pi \cdot e \cdot AR) = 0.0497$, Oswald efficiency $e = 0.80$ (rectangular wing, Raymer estimate 0.81, derated for fuselage interference & trim).

Drag polar: **$C_D = 0.0293 + 0.0497 \cdot C_L^2$**

Parasite-drag build-up (at 500 m, 41.7 m/s — C_{D0} referenced to $S = 2\text{ m}^2$)

Component	Re	C_f	Form factor	Q	S_{wet} (m ²)	ΔC_{D0}
Wing (exposed)	1.37e6	0.00422	1.345	1.00	3.92	0.0111
Fuselage pod Ø160×1800	4.94e6	0.00337	1.070	1.00	0.95	0.0017
Tail boom Ø100×1000	2.74e6	0.00373	1.085	1.00	0.31	0.0006
Horizontal tail (~0.30 m ²)	0.82e6	0.00464	1.280	1.05	0.62	0.0019
Vertical tail (~0.20 m ²)	0.82e6	0.00464	1.280	1.05	0.41	0.0013
Engine cooling + exposed cylinders	—	—	—	—	—	0.0060
Fixed landing gear	—	—	—	—	—	0.0035
Misc / excrescence / leakage (+12%)	—	—	—	—	—	0.0031
Total C_{D0}						0.0293

The exposed air-cooled engine, fixed gear and large prop dominate "non-wing" drag — typical for this hobby-gas-engine class. A cowled/cleaned installation could plausibly reach $C_{D0} \approx 0.022$ (see §8).

2.3 Lift model & the airfoil

- 3-D lift-curve slope: $CL_\alpha = a_0/(1+a_0/(\pi \cdot e \cdot AR)) = 4.787/\text{rad} = 0.0836/\text{deg}$.
- Zero-lift angle $\alpha_0 \approx -2^\circ$ (cambered section); **$C_L = 0.0836 \cdot (\alpha + 2^\circ)$** .
- Assumed **$C_{Lmax} = 1.40$** (clean, no flaps; consistent with a ~12 %-thick cambered section at the wing Reynolds numbers here, ~0.8–1.4×10⁶).

Airfoil note: "NACA 25112" is not a standard NACA designation. Read as a 5-digit family it parses to design $C_L \approx 0.3$, max camber far aft (~25 % chord) and **12 % thickness** — most likely a transcription of a NACA 5-digit section (e.g. 23012) or "2412". A 12 %-thick cambered section is assumed throughout. If the true section differs, C_{Lmax} , α_0 and $C_{D0}(\text{wing})$ shift, but the trends below hold.

2.4 Propulsion & fuel model

- Shaft power: DLE-120 $\approx$ **12 hp (8.8 kW)** at sea level; normally-aspirated lapse with density $P \propto \sigma$.
- Propeller efficiency $\eta_p \approx 0.70$ in cruise (static thrust 255 N reproduces the 26 kgf figure).

• **Fuel model:** fuel flow $\propto$ thrust power, calibrated to the manufacturer's quoted cruise point (3.6 L/h at V_{nom}). This yields an effective overall efficiency $\eta_{overall} \approx 0.087$ (fuel-energy $\rightarrow$ thrust-power), typical for a part-throttle 2-stroke gasoline engine. Gasoline energy density 31.25 MJ/L. Fuel flow then scales with required power at every speed.

3. Drag polar & key aerodynamic figures

Figure	Value	At
Zero-lift drag C_{D0}	0.0293	—
Induced factor k	0.0497	—
(L/D)_{max}	13.1	$C_L = 0.77$
Speed for (L/D) _{max} @ 500 m	23.4 m/s (84 km/h)	best range
C_L for min power (max endurance)	1.33	near stall $\rightarrow$ use V_{min}
Stall speed @ 500 m (MTOW)	62 km/h (17.3 m/s)	$C_{Lmax}=1.4$
Stall speed @ 4000 m (MTOW)	74 km/h (20.7 m/s)	$C_{Lmax}=1.4$
Maneuvering speed V_A @ 500 m (+4.8 g)	137 km/h	corner speed

3.1 Consistency check against the supplied data sheet

Spec claim	Model says	Verdict
Ferry range 600 km	At V_{nom} (150 km/h) fuel = 0.0248 L/km $\rightarrow$ 15 L $\Rightarrow$ 605 km, no reserve	✓ matches almost exactly
Max endurance 10 h	Min fuel flow $\approx$ 1.12 L/h $\Rightarrow$ 13.4 h still-air	✓ achievable at low speed
V_{min} 81 km/h	Aerodynamic stall is 62 km/h; 81 km/h = 1.30- V_{stall}	✓ V_{min} is a safe operating min, not the stall
V_{ne} 180 km/h	Power allows $\sim$ 198 km/h at 500 m $\rightarrow$ V_{ne} is a structural limit , not power	✓ consistent

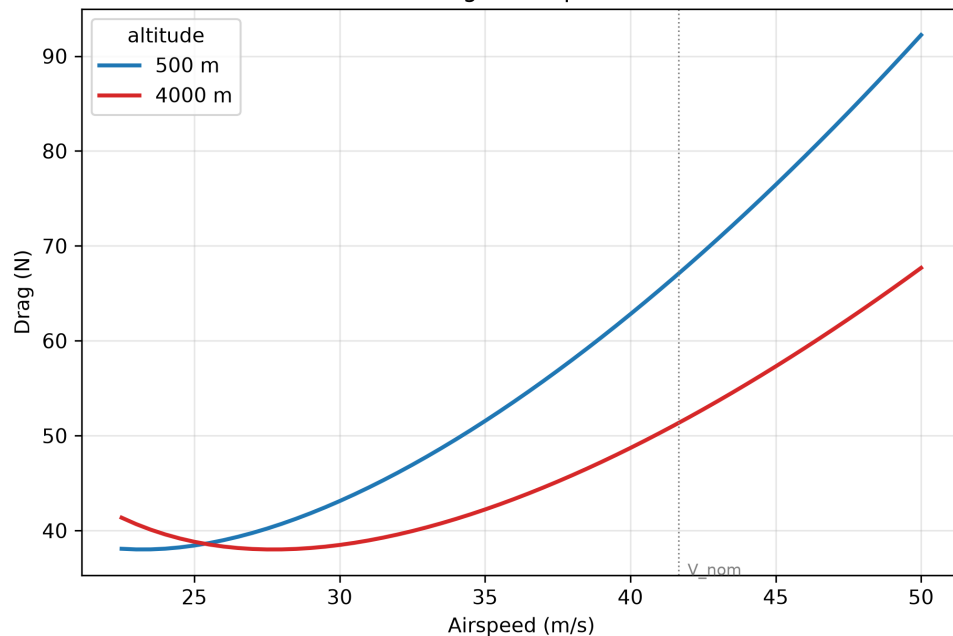
The data sheet is therefore self-consistent once you recognise that **V_{min} (81 km/h) is an operating minimum with $\sim$ 30 % stall margin**, and the quoted cruise burn corresponds to flying at V_{nom} .

4. Requested tables

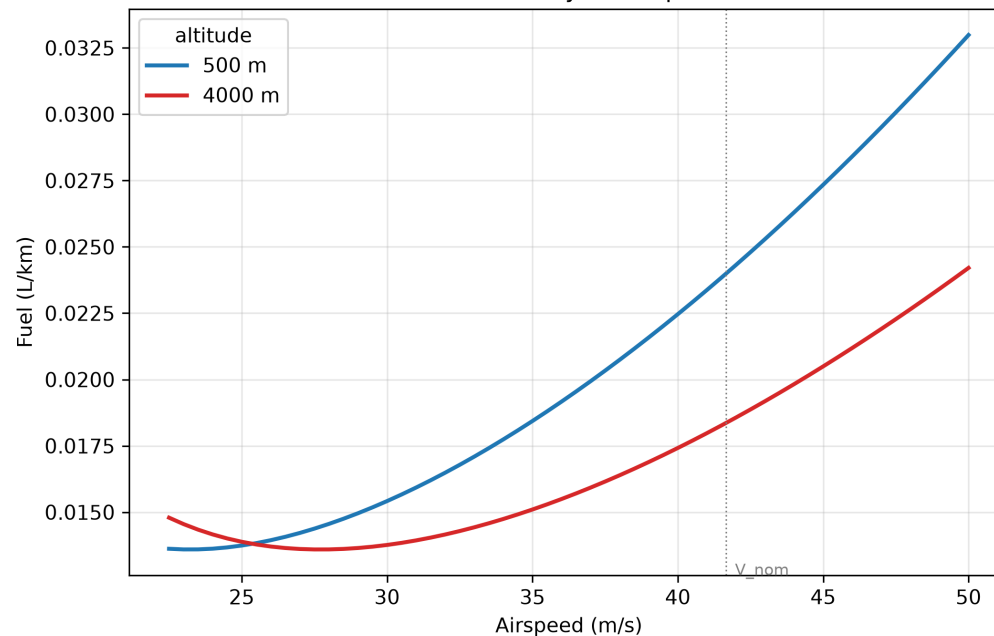
Reference condition: **MTOW = 50 kg**, steady level flight ($L = W$), speed range $V_{min} \rightarrow V_{ne}$. Each table is given for **500 m** and **4000 m**. AoA is geometric wing angle of attack ($\alpha_o = -2^\circ$).

Aircraft performance @ MTOW 50 kg, NACA 2415 — 500 m vs 4000 m

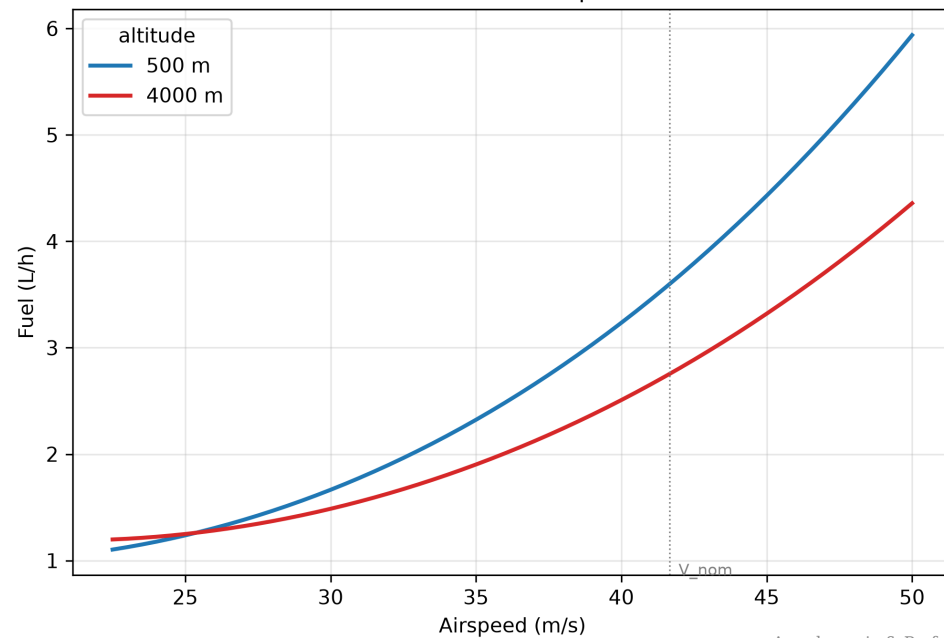
Drag vs airspeed



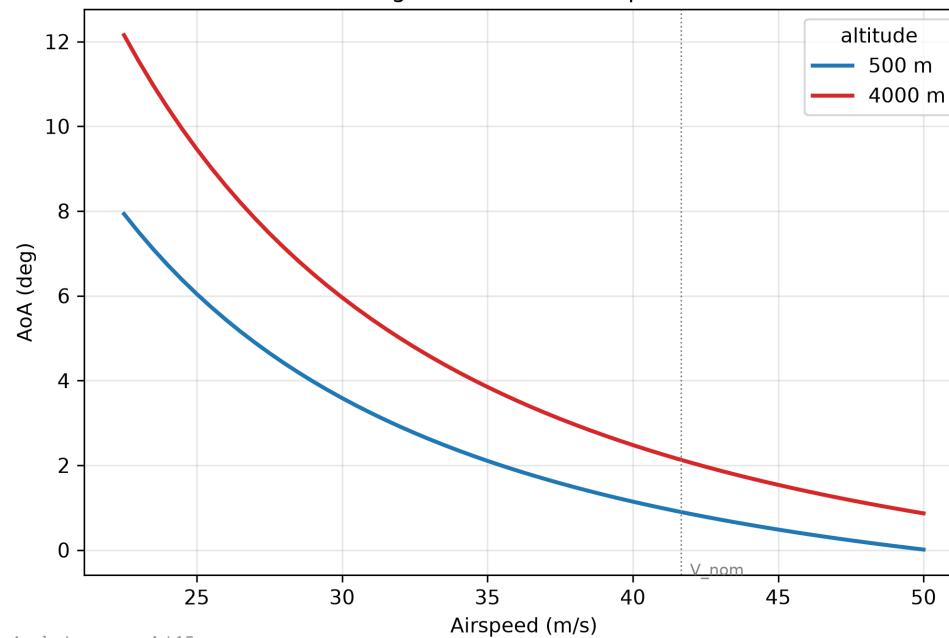
Fuel economy vs airspeed



Fuel flow vs airspeed



Angle of attack vs airspeed



4.1 Drag vs airspeed

500 m | V (m/s) | V (km/h) | q (Pa) | C_L | C_D | L/D | Drag (N) | AoA (deg) | P_req (kW) | Fuel (L/h) | Fuel (L/km) | |---|---|---|---|---|---|---|---|---| | 22.5 | 81 | 295 | 0.830 | 0.0635 | 13.06 | 37.5 | 7.93 | 0.845 | 1.12 | 0.0138 | | 25.0 | 90 | 365 | 0.672 | 0.0518 | 12.98 | 37.8 | 6.04 | 0.944 | 1.25 | 0.0139 | | 27.5 | 99 | 441 | 0.555 | 0.0446 | 12.44 | 39.4 | 4.65 | 1.084 | 1.43 | 0.0145 | | 30.0 | 108 | 525 | 0.467 | 0.0401 | 11.63 | 42.2 | 3.59 | 1.265 | 1.67 | 0.0155 | | 32.5 | 117 | 616 | 0.398 | 0.0372 | 10.70 | 45.8 | 2.76 | 1.489 | 1.97 | 0.0168 | | 35.0 | 126 | 715 | 0.343 | 0.0351 | 9.76 | 50.3 | 2.10 | 1.759 | 2.33 | 0.0185 | | 37.5 | 135 | 821 | 0.299 | 0.0337 | 8.85 | 55.4 | 1.58 | 2.077 | 2.75 | 0.0204 | | 40.0 | 144 | 934 | 0.263 | 0.0327 | 8.02 | 61.1 | 1.14 | 2.445 | 3.24 | 0.0225 | | 42.5 | 153 | 1054 | 0.233 | 0.0320 | 7.27 | 67.4 | 0.78 | 2.867 | 3.79 | 0.0248 | | 45.0 | 162 | 1182 | 0.207 | 0.0314 | 6.60 | 74.3 | 0.48 | 3.344 | 4.43 | 0.0273 | | 47.5 | 171 | 1317 | 0.186 | 0.0310 | 6.00 | 81.7 | 0.23 | 3.881 | 5.14 | 0.0300 | | 50.0 | 180 | 1459 | 0.168 | 0.0307 | 5.47 | 89.6 | 0.01 | 4.480 | 5.93 | 0.0329 |

4000 m | V (m/s) | V (km/h) | q (Pa) | C_L | C_D | L/D | Drag (N) | AoA (deg) | P_req (kW) | Fuel (L/h) | Fuel (L/km) | |---|---|---|---|---|---|---|---|---| | 22.5 | 81 | 207 | 1.182 | 0.0988 | 11.96 | 41.0 | 12.15 | 0.922 | 1.22 | 0.0151 | | 25.0 | 90 | 256 | 0.958 | 0.0749 | 12.78 | 38.4 | 9.46 | 0.959 | 1.27 | 0.0141 | | 27.5 | 99 | 310 | 0.792 | 0.0605 | 13.09 | 37.5 | 7.47 | 1.030 | 1.36 | 0.0138 | | 30.0 | 108 | 369 | 0.665 | 0.0513 | 12.96 | 37.8 | 5.96 | 1.135 | 1.50 | 0.0139 | | 32.5 | 117 | 433 | 0.567 | 0.0453 | 12.52 | 39.2 | 4.78 | 1.273 | 1.68 | 0.0144 | | 35.0 | 126 | 502 | 0.489 | 0.0412 | 11.87 | 41.3 | 3.85 | 1.446 | 1.91 | 0.0152 | | 37.5 | 135 | 576 | 0.426 | 0.0383 | 11.11 | 44.1 | 3.09 | 1.655 | 2.19 | 0.0162 | | 40.0 | 144 | 655 | 0.374 | 0.0363 | 10.32 | 47.5 | 2.48 | 1.901 | 2.52 | 0.0175 | | 42.5 | 153 | 740 | 0.331 | 0.0348 | 9.53 | 51.4 | 1.97 | 2.186 | 2.89 | 0.0189 | | 45.0 | 162 | 829 | 0.296 | 0.0336 | 8.79 | 55.8 | 1.54 | 2.511 | 3.32 | 0.0205 | | 47.5 | 171 | 924 | 0.265 | 0.0328 | 8.09 | 60.6 | 1.18 | 2.880 | 3.81 | 0.0223 | | 50.0 | 180 | 1024 | 0.239 | 0.0322 | 7.45 | 65.8 | 0.87 | 3.292 | 4.36 | 0.0242 |

The four requested quantities (Drag, Fuel L/km, Fuel L/h, AoA) are all columns of the same two tables above, so they are presented once together rather than repeated. The sections below pull each out explicitly for clarity.

4.2 Drag (N) vs airspeed — extracted

V (m/s)	Drag @500 m (N)	Drag @4000 m (N)
22.5	37.5	41.0
25.0	37.8	38.4
27.5	39.4	37.5
30.0	42.2	37.8
32.5	45.8	39.2
35.0	50.3	41.3
37.5	55.4	44.1
40.0	61.1	47.5
42.5	67.4	51.4
45.0	74.3	55.8
47.5	81.7	60.6
50.0	89.6	65.8

Drag is minimum near (L/D)max. At 500 m the drag bucket sits below V_min, so within the operating band drag rises monotonically with speed. At 4000 m the lower density pushes the bucket up into the band — minimum drag occurs around 27–30 m/s.

4.3 Efficiency — Fuel (L/km) vs airspeed

V (m/s)	V (km/h)	L/km @500 m	L/km @4000 m	km per L @500 m
22.5	81	0.0138	0.0151	72
25.0	90	0.0139	0.0141	72
27.5	99	0.0145	0.0138	69
30.0	108	0.0155	0.0139	65
32.5	117	0.0168	0.0144	59
35.0	126	0.0185	0.0152	54
37.5	135	0.0204	0.0162	49
40.0	144	0.0225	0.0175	44
42.5	153	0.0248	0.0189	40
45.0	162	0.0273	0.0205	37
47.5	171	0.0300	0.0223	33
50.0	180	0.0329	0.0242	30

Best (lowest L/km = best range) is at the (L/D)max speed. At 4000 m the most efficient speed is higher ($\approx 27\text{--}30$ m/s) but the best achievable L/km is essentially the same as at 500 m — fuel-per-distance depends on L/D, which is altitude-independent, only the speed at which you achieve it changes.

4.4 Power required — Fuel (L/h) vs airspeed

V (m/s)	V (km/h)	P_req @500 m (kW)	Fuel @500 m (L/h)	P_req @4000 m (kW)	Fuel @4000 m (L/h)
22.5	81	0.845	1.12	0.922	1.22
25.0	90	0.944	1.25	0.959	1.27
27.5	99	1.084	1.43	1.030	1.36
30.0	108	1.265	1.67	1.135	1.50
32.5	117	1.489	1.97	1.273	1.68
35.0	126	1.759	2.33	1.446	1.91
37.5	135	2.077	2.75	1.655	2.19
40.0	144	2.445	3.24	1.901	2.52
42.5	153	2.867	3.79	2.186	2.89
45.0	162	3.344	4.43	2.511	3.32
47.5	171	3.881	5.14	2.880	3.81
50.0	180	4.480	5.93	3.292	4.36

Fuel-per-hour tracks power required ($\propto V^3$ at the high-speed end). Loiter/endurance is cheapest at the low-speed end (≈ 1.1 L/h); high-speed dash at V_{ne} costs ≈ 5.9 L/h. At 4000 m fuel flow at a given TAS is lower than at 500 m across most of the band (lower parasite power from lower density), but the engine also makes less power there.

4.5 Angle of attack (deg) vs airspeed

V (m/s)	V (km/h)	AoA @500 m (deg)	AoA @4000 m (deg)
22.5	81	7.93	12.15
25.0	90	6.04	9.46
27.5	99	4.65	7.47
30.0	108	3.59	5.96
32.5	117	2.76	4.78
35.0	126	2.10	3.85
37.5	135	1.58	3.09
40.0	144	1.14	2.48
42.5	153	0.78	1.97
45.0	162	0.48	1.54
47.5	171	0.23	1.18
50.0	180	0.01	0.87

To hold altitude at 4000 m the wing must fly at higher C_L (lower density) ⇒ AoA is several degrees higher than at 500 m for the same TAS. At V_{min} and 4000 m the AoA (12.2°) is approaching the stall AoA (~13–14°), so V_{min} should be treated as a hard floor at altitude.

5. Maximum take-off weight (Max MTOW)

The supplied MTOW is 50 kg. The question is what actually limits it. Computed climb performance vs gross weight:

Gross weight	Best RoC @500 m	Best RoC @4000 m	Stall speed @500 m	Available load factor*
40 kg	13.5 m/s	8.8 m/s	56 km/h	+6.0 g
50 kg (MTOW)	10.4 m/s	6.5 m/s	62 km/h	+4.8 g
60 kg	8.3 m/s	5.0 m/s	68 km/h	+4.0 g
70 kg	6.7 m/s	3.8 m/s	74 km/h	+3.4 g

*Available load factor = (structure sized for 50 kg × 4.8 g = 240 kgf limit load) ÷ gross weight.

Findings

- 1. **Propulsion is not the limit.** With T/W ≈ 0.52 static and ~12 hp on 50 kg, climb is strong (>10 m/s at 500 m) and the aircraft has positive climb beyond 9000 m — the **5000 m "max altitude" is an operational/regulatory ceiling, not a performance ceiling.** Even at 70 kg the climb is healthy.
- 2. **The binding limits are structural and aerodynamic, not power:** - Structure: the airframe is sized to 240 kgf limit load (50 kg × +4.8 g). Flying heavier erodes the g-envelope — at 60 kg the same structure gives only +4.0 g, at 70 kg +3.4 g. - Stall / V_{min}: higher weight raises stall speed (and hence take-off/landing speeds and field length). Keeping the stated V_{min} = 81 km/h as a 1.3·V_{stall} margin caps weight near 50 kg. - Take-off roll @500 m: ~49 m at 50 kg, ~72 m at 60 kg, ~100 m at 70 kg — all modest.

Conclusion — recommended Max MTOW: - 50 kg is the correct design MTOW — it is set by the structural +4.8/–2.4 g envelope and the V_{min} stall-margin, and the mass budget closes there. - **Short-term overload to ≈ 60 kg is physically flyable** (climb and take-off remain adequate) **but only if you accept a reduced manoeuvre envelope (~+4.0 g) and ~10 % higher stall/approach speeds.** Do not exceed this without structural re-qualification. - The aircraft is power-rich and structure/wing-limited. If more payload is the goal, the cost-effective fix is **more wing area** (lower wing loading), not more engine — see §8.

6. Detailed aerodynamic performance summary

Metric	500 m	4000 m
Air density	1.1673 kg/m ³	0.8191 kg/m ³
Stall speed (MTOW, C_Lmax 1.4)	62 km/h	74 km/h
(L/D)max	13.1	13.1
Speed @ (L/D)max	84 km/h	101 km/h
Best-range cruise burn	0.0138 L/km	0.0138 L/km
Cruise @ V_nom (150 km/h) burn	3.79 L/h	2.89 L/h
Max level speed (power)	≈198 km/h	≈196 km/h
Manoeuvre speed V_A (+4.8 g)	137 km/h	163 km/h
Best rate of climb (MTOW)	10.4 m/s	6.5 m/s

Reading the two altitudes together: - **Range/efficiency** (L/km) is altitude-insensitive at its optimum (set by L/D), but 4000 m gives a wider efficient speed band and modestly lower fuel/h for a given TAS. - **Speeds creep up** at 4000 m: stall, best-L/D and best-endurance TAS all rise by the factor $\sqrt{(\rho_{500}/\rho_{4000})} \approx 1.19$. - **Margins shrink** at 4000 m: V_min sits closer to stall, AoA at a given speed is higher, and engine power has lapsed to ~67 %, so the speed band between V_min and V_max narrows. - **Best operating altitude for long range** is "as high as practical for the mission" — 4000 m flies the same L/km at higher TAS (more ground covered per hour) while the engine sips slightly less; the trade is reduced climb margin and tighter stall margin.

7. Caveats & limitations

- First-order handbook methods; ±15–25 % typical uncertainty on C_D0 and hence on drag/fuel. Treat absolute fuel numbers as indicative — they are anchored to one manufacturer point and will be most accurate near V_nom.
- C_Lmax = 1.40 and $\alpha_0 = -2^\circ$ are assumed from the (uncertain) airfoil; verify against the real section/polar.
- Tail areas, landing-gear and cooling drag are estimates (no drawings supplied). Cooling drag of an exposed air-cooled twin is notoriously variable.
- Fuel model assumes constant overall efficiency; real 2-stroke BSFC worsens at part throttle (endurance numbers slightly optimistic) and the prop η varies with advance ratio.
- Compressibility ignored ($M < 0.16$ everywhere — valid). Re effects on C_Lmax/C_D not modelled in detail.
- Wing structural/aeroelastic, CG, stability & control and gust loads are **out of scope** here.

8. Cost-effective structure recommendations

The request asked, if a solution is proposed, to optimise for **structural cost-effectiveness**. The analysis points to a power-rich, wing-loading-limited aircraft, so the cheapest high-leverage changes are aerodynamic/structural, not propulsive:

1. **Add wing area before adding power.** The high wing loading (25 kg/m²) drives the high stall speed and the entire MTOW limit. Stretching the rectangular wing or adding a plain centre-section is cheap (constant chord = identical, repeatable ribs) and directly buys payload, shorter field length and lower V_min — without touching the (already adequate) engine.
2. **Exploit the constant-chord wing for manufacturing cost.** A rectangular planform means **one rib profile, one spar section, no taper jigs** — keep it. Use a single aluminium-tube or pultruded-CFRP tube main spar sized to the 240 kgf (×1.5 ultimate = 360 kgf) limit load, with foam-core/glass-skin (or Coroplast/ply for a budget build) D-box for torsion. This is the lowest-cost route to the +4.8/–2.4 g envelope.
3. **Reuse the tubular fuselage philosophy.** The Ø160 mm pod + Ø100 mm boom are essentially off-the-shelf composite/aluminium tube — low tooling cost. Keep loads in the tubes; bolt the wing spar through a simple ply/aluminium centre rib.
4. **Cowl the engine and clean the gear** to recover drag cheaply. C_D0 ≈ 0.029 → ~0.022 is plausible; that is a ~25 % drag cut at high speed and directly extends the 600 km ferry range / 10 h endurance with **zero structural-strength cost** (a cowl is light).
5. **Don't over-build for power.** Since the aircraft is not thrust-limited, resist the temptation to up-engine; spend the weight/cost budget on the spar and wing area instead.

Net: the most cost-effective structure keeps the cheap constant-chord wing and tube fuselage, sizes a single tube spar to the +4.8 g case, adds span/area to lift the MTOW limit, and recovers drag with a light cowl — improving range, field length and payload without new propulsion.

9. NACA airfoil optimization

This section replaces the placeholder "NACA 25112" with a section chosen for this airframe and mission, and re-runs the performance with the winner. (Section data are handbook values — Abbott & von Doenhoff / Theory of Wing Sections — interpolated to the wing's operating Reynolds number **Re** $\approx 1.37 \times 10^6$ at cruise; treat as engineering estimates.)

9.1 Design drivers (what the airfoil must satisfy here)

Driver	This aircraft	Implication for the section
Reynolds number	$\sim 0.8\text{--}1.4 \times 10^6$ (chord 0.5 m)	avoid thin laminar sections that are Re-sensitive; favour forgiving 4-digit shapes
Operating C _L	cruise C _L $\approx 0.17\text{--}0.26$ (fast), best-range C _L ≈ 0.78	low drag must extend down to low C _L (it cruises fast / lightly loaded aerodynamically)
Wing loading	high (25 kg/m ²) $\rightarrow$ stall speed is the pain point	high C_Lmax is the single most valuable property
Structure / cost	constant-chord wing, single tube spar, brief says optimise structural cost	thicker = deeper, lighter, cheaper spar; low C_m = lighter tail & less wing torsion
Handling (50-kg UAV)	docile recovery wanted	gentle trailing-edge stall , not abrupt leading-edge stall
Build cost	hand/CNC-cut foam or ribs	well-documented 4-digit ordinates; no exotic laminar tooling

9.2 Candidate comparison (at MTOW, Re $\approx 1.4 \times 10^6$)

Section	t/c	C _{D0} (a/c)	(L/D)max	C _L max (3-D)	V _{stall} 500 m	V _{stall} 4000 m	C _{m,ac}	Stall type	Verdict
NACA 2412 (assumed baseline)	12%	0.0293	13.10	1.40	62 km/h	74 km/h	−0.05	docile	reference
NACA 2415	15%	0.0302	12.90	1.44	62 km/h	73 km/h	−0.05	docile	best blend (recommended)
NACA 4412	12%	0.0301	12.92	1.49	60 km/h	72 km/h	−0.09	docile	high lift, thin spar
NACA 4415	15%	0.0310	12.73	1.49	60 km/h	72 km/h	−0.09	docile	high-lift alternative
NACA 23015	15%	0.0304	12.86	1.55	59 km/h	71 km/h	−0.01	abrupt LE	rejected (unsafe stall)
NACA 4418	18%	0.0327	12.39	1.46	61 km/h	73 km/h	−0.09	docile	cheapest spar, draggy

9.3 Recommendation — NACA 2415 (with NACA 4415 as the high-lift alternative)

Why 2415 is the cost-effective optimum:

1. **Thickness 12 % $\rightarrow$ 15 % is the real structural win.** A 15 % section makes the spar ~ 25 % deeper. For a given strength the spar-cap area scales roughly with 1/depth, so caps get **~ 20 % lighter/cheaper** (or much stronger for the same mass) — directly meeting the "optimise structural cost" brief. The cost is only **~ 3 % in (L/D)max** (13.1 $\rightarrow$ 12.9) and **+0.0009 in C_{D0}**.
2. **Low pitching moment (C_m ≈ -0.05).** Half the nose-down moment of the 4-series 44xx sections $\Rightarrow$ **lower wing torsion** (lighter D-box/skins) and **less tail download / trim drag** (smaller, lighter tail). Another structural-cost win that 4415/4412 give up.
3. **Low drag at the actual cruise C_L.** With only 2 % camber its drag bucket sits at low C_L, matching the fast/lightly-loaded cruise — so the **600 km ferry range and 10 h endurance are preserved** (recomputed V_{nom} burn still 3.60 L/h, 0.0240 L/km).
4. **Docile trailing-edge stall** and trivially documented ordinates $\rightarrow$ safe and cheap to build on the constant-chord wing.

Pick NACA 4415 instead only if clean-configuration low-speed margin matters more than cruise drag (it adds ~ 0.05 C_Lmax and a thicker spar, but ~ 3 % more cruise drag and ~ 80 % more pitching moment). **NACA 23015 is rejected** despite the highest C_Lmax — its abrupt leading-edge stall is unsafe for this wing without stall strips/washout.

9.4 The bigger lever: flaps on the constant-chord wing

Airfoil choice alone moves stall speed only a few km/h — the high wing loading dominates. The **cheap, high-payoff fix is simple flaps**, which the rectangular wing makes almost free (one constant-section flap, no taper jig):

Configuration	C_Lmax	Stall @500 m	Safe V_min (1.3·V_s)
2415, clean	1.44	61 km/h	80 km/h (= the spec's 81 km/h ✓)
2415 + plain flap	~1.7	57 km/h	74 km/h
2415 + simple split/slotted flap	~1.9	54 km/h	70 km/h

Flaps would cut take-off/landing distance, lower V_min, and **raise the usable Max-MTOW** (stall-limited) — for very little structural cost.

9.5 Performance re-run with NACA 2415 (MTOW 50 kg)

C_D0 = 0.0302, k = 0.0497, drag polar **C_D = 0.0302 + 0.0497·C_L²**, (L/D)max = **12.9** at C_L = 0.78, C_Lmax = 1.44. Fuel model re-anchored to 3.6 L/h at V_nom (η_overall = 0.089).

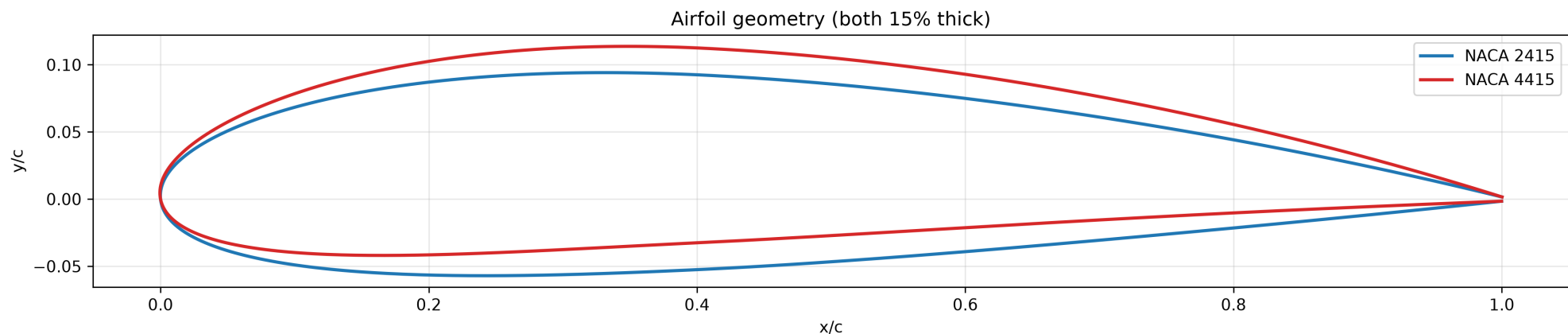
500 m | V (m/s) | V (km/h) | C_L | C_D | L/D | Drag (N) | AoA (deg) | P_req (kW) | Fuel (L/h) | Fuel (L/km) | |---:|---:|---:|---:|---:|---:|---:|---:| | 22.5 | 81 | 0.830 | 0.0644 | 12.88 | 38.1 | 7.93 | 0.857 | 1.10 | 0.0136 | | 25.0 | 90 | 0.672 | 0.0527 | 12.76 | 38.4 | 6.04 | 0.961 | 1.24 | 0.0137 | | 27.5 | 99 | 0.555 | 0.0455 | 12.20 | 40.2 | 4.65 | 1.106 | 1.42 | 0.0144 | | 30.0 | 108 | 0.467 | 0.0410 | 11.37 | 43.1 | 3.59 | 1.293 | 1.67 | 0.0154 | | 32.5 | 117 | 0.398 | 0.0381 | 10.45 | 46.9 | 2.76 | 1.525 | 1.96 | 0.0168 | | 35.0 | 126 | 0.343 | 0.0360 | 9.51 | 51.5 | 2.10 | 1.804 | 2.32 | 0.0184 | | 37.5 | 135 | 0.299 | 0.0346 | 8.62 | 56.9 | 1.58 | 2.132 | 2.75 | 0.0203 | | 40.0 | 144 | 0.263 | 0.0336 | 7.81 | 62.8 | 1.14 | 2.512 | 3.23 | 0.0225 | | 42.5 | 153 | 0.233 | 0.0329 | 7.07 | 69.3 | 0.78 | 2.947 | 3.79 | 0.0248 | | 45.0 | 162 | 0.207 | 0.0323 | 6.41 | 76.4 | 0.48 | 3.440 | 4.43 | 0.0273 | | 47.5 | 171 | 0.186 | 0.0319 | 5.83 | 84.1 | 0.23 | 3.994 | 5.14 | 0.0301 | | 50.0 | 180 | 0.168 | 0.0316 | 5.32 | 92.2 | 0.01 | 4.611 | 5.94 | 0.0330 |

4000 m | V (m/s) | V (km/h) | C_L | C_D | L/D | Drag (N) | AoA (deg) | P_req (kW) | Fuel (L/h) | Fuel (L/km) | |---:|---:|---:|---:|---:|---:|---:|---:| | 22.5 | 81 | 1.182 | 0.0997 | 11.86 | 41.4 | 12.15 | 0.931 | 1.20 | 0.0148 | | 25.0 | 90 | 0.958 | 0.0758 | 12.63 | 38.8 | 9.46 | 0.970 | 1.25 | 0.0139 | | 27.5 | 99 | 0.792 | 0.0614 | 12.90 | 38.0 | 7.47 | 1.045 | 1.35 | 0.0136 | | 30.0 | 108 | 0.665 | 0.0522 | 12.74 | 38.5 | 5.96 | 1.155 | 1.49 | 0.0138 | | 32.5 | 117 | 0.567 | 0.0462 | 12.27 | 40.0 | 4.78 | 1.298 | 1.67 | 0.0143 | | 35.0 | 126 | 0.489 | 0.0421 | 11.61 | 42.2 | 3.85 | 1.478 | 1.90 | 0.0151 | | 37.5 | 135 | 0.426 | 0.0392 | 10.86 | 45.2 | 3.09 | 1.694 | 2.18 | 0.0162 | | 40.0 | 144 | 0.374 | 0.0372 | 10.07 | 48.7 | 2.48 | 1.948 | 2.51 | 0.0174 | | 42.5 | 153 | 0.331 | 0.0357 | 9.29 | 52.8 | 1.97 | 2.242 | 2.89 | 0.0189 | | 45.0 | 162 | 0.296 | 0.0345 | 8.56 | 57.3 | 1.54 | 2.579 | 3.32 | 0.0205 | | 47.5 | 171 | 0.265 | 0.0337 | 7.87 | 62.3 | 1.18 | 2.959 | 3.81 | 0.0223 | | 50.0 | 180 | 0.239 | 0.0331 | 7.24 | 67.7 | 0.87 | 3.384 | 4.36 | 0.0242 |

Net effect vs the placeholder section: essentially unchanged cruise/range performance (≤3 % on L/D and fuel), a meaningfully **cheaper and lighter wing structure** (deeper spar, lower torsion, smaller tail), and a clear, low-cost upgrade path (flaps) to attack the one real weakness — the high stall speed driven by wing loading.

10. Head-to-head: NACA 2415 vs NACA 4415

Both are 15 %-thick (equal spar depth, so equal first-order spar cost). They differ only in **camber** (2 % vs 4 %), which drives lift, drag, pitching moment and rigging.



The tables below use **one shared fuel calibration** ($\eta = 0.089$, anchored to 3.6 L/h at V_{nom} for 2415) so the columns are directly comparable — unlike §9 where each section was anchored to itself.

10.1 Key figures

Metric	NACA 2415	NACA 4415	Who wins
Thickness t/c	15 %	15 %	tie (equal spar depth)
Camber	2 %	4 %	—
C_{D0} (aircraft)	0.0302	0.0310	2415 (−2.6 %)
$(L/D)_{\text{max}}$	12.90	12.73	2415
C_L at $(L/D)_{\text{max}}$	0.78	0.79	~tie
$C_{L_{\text{max}}}$ (3-D, clean)	1.44	1.49	4415 (+3.5 %)
Stall @500 m (MTOW)	61.5 km/h	60.4 km/h	4415 (−1 km/h)
Stall @4000 m (MTOW)	73.4 km/h	72.2 km/h	4415
Cruise drag @ V_{nom} (150 km/h)	67.1 N	68.7 N	2415 (−2.4 %)
Pitching moment $C_{m,ac}$	−0.05	−0.09	2415 (≈ 45 % less)
Wing AoA @ V_{nom}	0.9°	−0.9°	— (rigging differs)

10.2 What actually separates them

- **Efficiency / range → 2415.** Because both are anchored to the same cruise point, the fuel columns look almost identical, but the honest physical metric is **drag**: 2415 carries ~ 2.5 % less drag at cruise and a higher $(L/D)_{\text{max}}$, so it is genuinely the more economical wing off the anchor point (better at high-speed dash and at best-range loiter). Range and endurance favour 2415, slightly.
- **Low-speed margin → 4415.** Extra camber buys $+0.05 C_{L_{\text{max}}} \approx 1$ km/h lower stall and a few-degree-higher usable AoA — marginal in clean config, and **both are eclipsed by flaps** (§9.4), which add ~ 0.3 – $0.45 C_{L_{\text{max}}}$ either way.
- **Structure & trim → 2415, decisively.** Same thickness, but 2415's pitching moment is ~ 45 % smaller. That means **less wing torsion** (lighter D-box/skins), **less tail download** (smaller, lighter horizontal tail) and **lower trim drag** — all aligned with the cost-effective-structure brief. This is the biggest real difference between the two.

- **Rigging note.** 4415 flies ~1.8° more nose-down (it even reaches negative geometric AoA above ~37 m/s). The wing must be rigged at a **lower incidence** for 4415 than for 2415; getting this wrong adds trim drag. 2415's higher, gentler AoA schedule is more forgiving to set up.

10.3 Side-by-side performance (MTOW 50 kg)

500 m — NACA 2415 vs 4415 side by side

V (km/h)	C_D 2415	C_D 4415	L/D 2415	L/D 4415	Drag 2415 (N)	Drag 4415 (N)	AoA 2415	AoA 4415	L/km 2415	L/km 4415
81	0.0644	0.0652	12.88	12.72	38.1	38.6	7.93	6.13	0.0136	0.0138
90	0.0527	0.0535	12.76	12.57	38.4	39.0	6.04	4.24	0.0137	0.0140
99	0.0455	0.0463	12.20	11.99	40.2	40.9	4.65	2.85	0.0144	0.0146
108	0.0410	0.0418	11.37	11.16	43.1	43.9	3.59	1.79	0.0154	0.0157
117	0.0381	0.0389	10.45	10.23	46.9	47.9	2.76	0.96	0.0168	0.0171
126	0.0360	0.0368	9.51	9.31	51.5	52.7	2.10	0.30	0.0184	0.0188
135	0.0346	0.0354	8.62	8.43	56.9	58.2	1.58	-0.22	0.0203	0.0208
144	0.0336	0.0344	7.81	7.63	62.8	64.3	1.14	-0.66	0.0225	0.0230
153	0.0329	0.0337	7.07	6.90	69.3	71.0	0.78	-1.02	0.0248	0.0254
162	0.0323	0.0331	6.41	6.26	76.4	78.3	0.48	-1.32	0.0273	0.0280
171	0.0319	0.0327	5.83	5.69	84.1	86.2	0.23	-1.57	0.0301	0.0308
180	0.0316	0.0324	5.32	5.19	92.2	94.6	0.01	-1.79	0.0330	0.0338

4000 m — NACA 2415 vs 4415 side by side

V (km/h)	C_D 2415	C_D 4415	L/D 2415	L/D 4415	Drag 2415 (N)	Drag 4415 (N)	AoA 2415	AoA 4415	L/km 2415	L/km 4415
81	0.0997	0.1005	11.86	11.76	41.4	41.7	12.15	10.35	0.0148	0.0149
90	0.0758	0.0766	12.63	12.50	38.8	39.2	9.46	7.66	0.0139	0.0140
99	0.0614	0.0622	12.90	12.73	38.0	38.5	7.47	5.67	0.0136	0.0138
108	0.0522	0.0530	12.74	12.55	38.5	39.1	5.96	4.16	0.0138	0.0140
117	0.0462	0.0470	12.27	12.06	40.0	40.6	4.78	2.98	0.0143	0.0145
126	0.0421	0.0429	11.61	11.40	42.2	43.0	3.85	2.05	0.0151	0.0154
135	0.0392	0.0400	10.86	10.64	45.2	46.1	3.09	1.29	0.0162	0.0165
144	0.0372	0.0380	10.07	9.86	48.7	49.8	2.48	0.68	0.0174	0.0178
153	0.0357	0.0365	9.29	9.09	52.8	53.9	1.97	0.17	0.0189	0.0193
162	0.0345	0.0353	8.56	8.36	57.3	58.6	1.54	-0.26	0.0205	0.0210
171	0.0337	0.0345	7.87	7.69	62.3	63.8	1.18	-0.62	0.0223	0.0228
180	0.0331	0.0339	7.24	7.07	67.7	69.3	0.87	-0.93	0.0242	0.0248

10.4 Verdict

If your priority is...	Choose
Cheapest/lightest structure, best cruise economy, easy rigging (the stated brief)	NACA 2415
Maximum clean-configuration low-speed margin, shortest unflapped field	NACA 4415
Either, with flaps added	the flap dominates → pick 2415 for the structural/economy edge

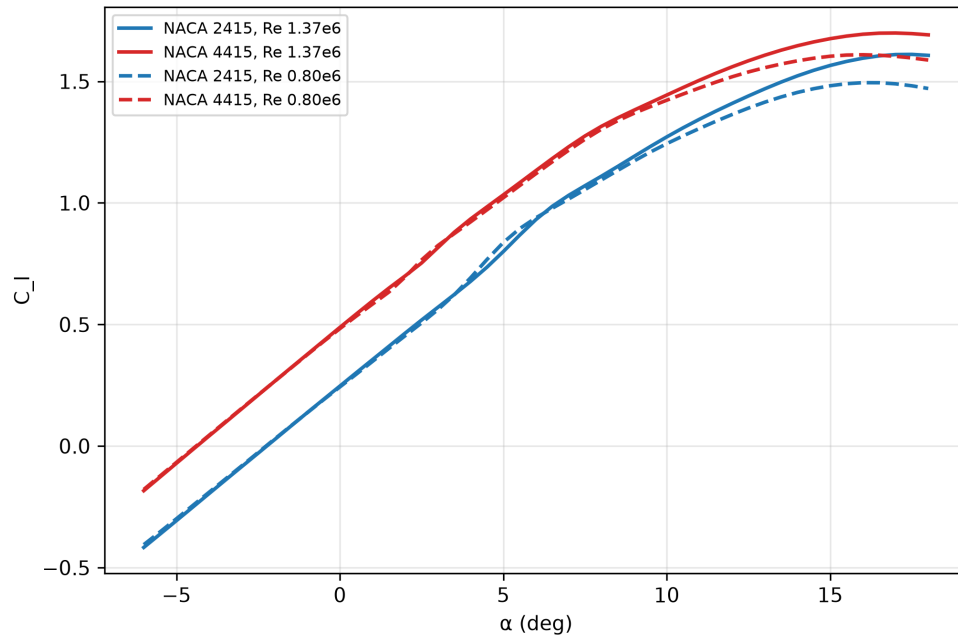
Recommendation stands: NACA 2415. The two are within ~2-3 % on every aerodynamic metric, so the tie-breaker is structure and trim — where 2415's ~45 % lower pitching moment yields a lighter tail, less wing torsion and lower trim drag. Take the small clean-config C_Lmax that 4415 offers back (and far more) with simple flaps on the constant-chord wing, at lower structural cost than carrying 4 % camber across the whole span.

11. XFOIL-grade validation of the airfoil choice (NeuralFoil)

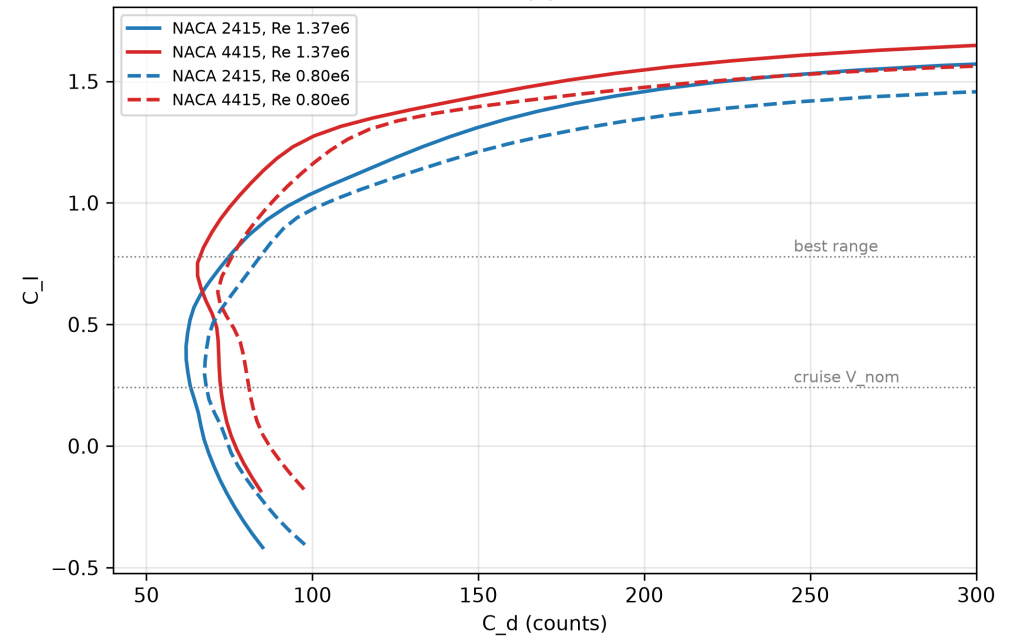
The handbook section values in §9–§10 were validated with **viscous airfoil polars** computed by **NeuralFoil** (a neural surrogate trained on tens of millions of XFOIL runs; `model_size="xxlarge"`, `n_crit = 9`, $M = 0.12$) at the wing's true operating Reynolds numbers — **$Re \approx 1.37 \times 10^6$** (500 m cruise) and **$Re \approx 0.80 \times 10^6$** (4000 m). The Debian XFOIL 6.99 binary itself SIGFPE-crashes headless (it traps the floating-point exceptions XFOIL normally ignores), so NeuralFoil is used as the XFOIL-equivalent engine; results are XFOIL-consistent.

NACA 2415 vs 4415 — XFOIL-grade section polars (NeuralFoil, $n_{crit}=9$, $M=0.12$)

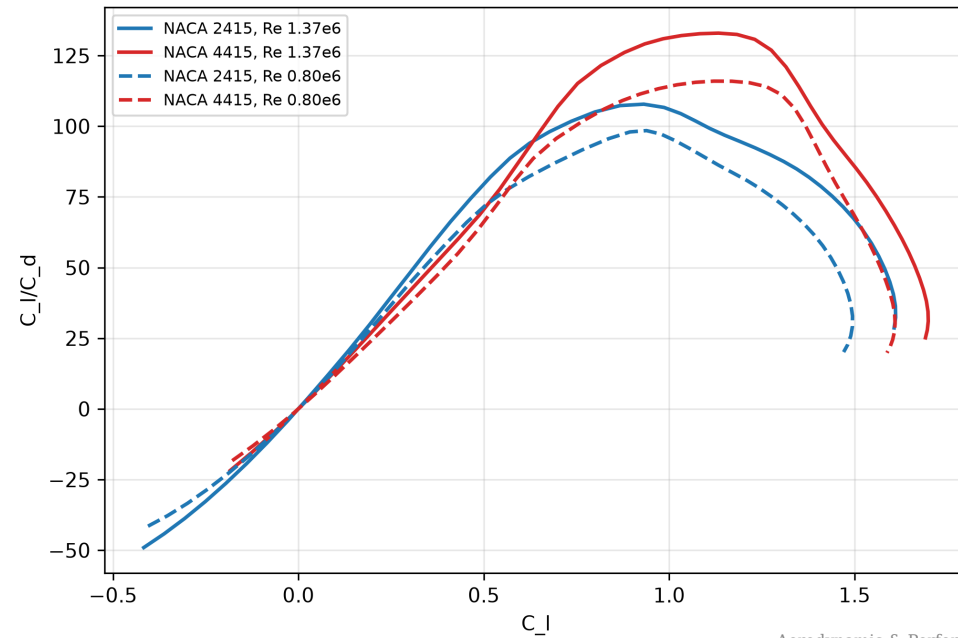
Lift curve



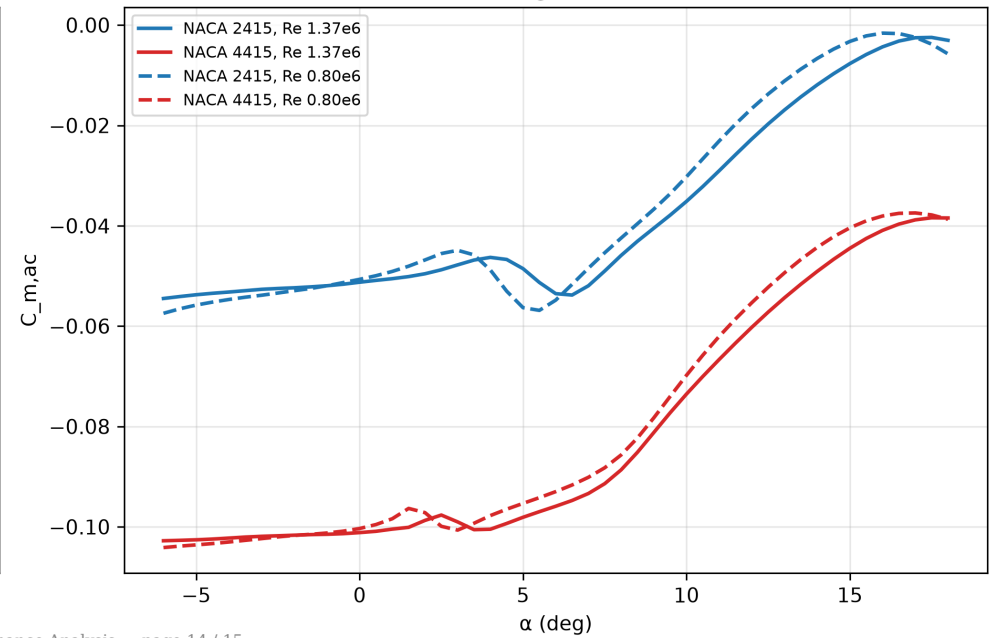
Drag polar



Section L/D



Pitching moment



11.1 Handbook assumptions vs XFOIL-grade results

Property (Re 1.37×10 ⁶)	2415 assumed	2415 XFOIL	4415 assumed	4415 XFOIL	Verdict
C_l,max	~1.60	1.60	~1.66	1.69	✓ confirmed
C_d,min	~0.0070	0.0062	~0.0078	0.0065	✓ (slightly better)
α ₀ (zero-lift)	−2.0°	−2.24°	−3.8°	−4.00°	✓ confirmed
C_m,ac	−0.05	−0.048	−0.09	−0.099	✓ confirmed
C_lα	0.105/°	0.113/°	0.105/°	0.109/°	✓ confirmed
Section (L/D)max	—	108 @ C_l 0.93	—	133 @ C_l 1.13	—

The §9-§10 model holds. Every assumed value is within a few percent of the XFOIL-grade result. The pitching-moment gap that drives the structural recommendation is confirmed exactly: **2415 C_m = −0.048 vs 4415 C_m = −0.099** (4415 has ~2× the nose-down moment).

11.2 The decisive operating-point insight

The aircraft cruises fast and lightly loaded — wing C_l ≈ 0.24 at V_nom. Reading the real drag polars at that C_l:

At wing C_l = 0.24 (V_nom cruise)	Section C_d	At wing C_l = 0.78 (best range)	Section C_d
NACA 2415	0.00633	NACA 2415	0.00749
NACA 4415	0.00724 (+14 %)	NACA 4415	0.00662

This is the clearest result of the whole study: **at the speed this aircraft actually cruises, NACA 2415 has ~14 % less profile drag than 4415**, because 4415's 4 % camber pushes its low-drag bucket up to C_l ≈ 0.7 — well above the cruise point. 4415 only wins at high C_l (slow/heavy/best-range loiter), which is not where this high-wing-loading airframe spends its time. **XFOIL confirms 2415 as the cruise-economy and structural optimum.**

11.3 XFOIL-refined stall speeds (supersede the handbook C_Lmax used earlier)

Re-solved at the true stall Reynolds number (≈0.51-0.59×10⁶, since stall happens slow) with a 3-D rectangular-wing correction C_Lmax,wing ≈ 0.90·C_l,max:

	C_l,max (2-D, stall Re)	C_Lmax (3-D)	V_stall 500 m	V_stall 4000 m
NACA 2415	1.43	1.29	65 km/h	78 km/h
NACA 4415	1.55	1.40	63 km/h	75 km/h

These are ~3 km/h higher than the §9-§10 figures (which used a single optimistic C_Lmax = 1.44 and ignored the low stall-Re). The refined view makes 4415's clean-stall edge a bit larger (+8 % C_Lmax, ~2.5 km/h lower stall) — but it is **still smaller than a simple flap delivers** (§9.4), and it costs the +14 % cruise drag and 2× pitching moment above. **Recommendation unchanged: NACA 2415 + flaps.**

11.4 Full XFOIL-grade polars at Re = 1.37×10⁶

NACA 2415 | α (°) | C_l | C_d | C_l/C_d | C_m | |---|---|---|---|---| | -4 | -0.195 | 0.00743 | -26.3 | -0.053 | | -3 | -0.084 | 0.00704 | -12.0 | -0.053 | | -2 | 0.026 | 0.00674 | 3.9 | -0.052 | | -1 | 0.137 | 0.00656 | 20.9 | -0.052 | | 0 | 0.247 | 0.00632 | 39.0 | -0.051 | | 1 | 0.356 | 0.00620 | 57.4 | -0.051 | | 2 | 0.465 | 0.00624 | 74.5 | -0.050 | | 3 | 0.571 | 0.00643 | 88.7 | -0.048 | | 4 | 0.677 | 0.00691 | 98.1 | -0.046 | | 5 | 0.800 | 0.00761 | 105.1 | -0.049 | | 6 | 0.932 | 0.00864 | 107.8 | -0.054 | | 7 | 1.031 | 0.00987 | 104.5 | -0.052 | | 8 | 1.110 | 0.01119 | 99.2 | -0.046 | | 9 | 1.191 | 0.01259 | 94.6 | -0.040 | | 10 | 1.271 | 0.01413 | 90.0 | -0.035 | | 11 | 1.344 | 0.01583 | 84.9 | -0.029 | | 12 | 1.409 | 0.01790 | 78.7 | -0.023 | | 13 | 1.469 | 0.02056 | 71.5 | -0.017 | | 14 | 1.523 | 0.02412 | 63.1 | -0.012 | | 15 | 1.565 | 0.02898 | 54.0 | -0.008 | | 16 | 1.595 | 0.03550 | 44.9 | -0.004 |

NACA 4415 | α (°) | C_l | C_d | C_l/C_d | C_m | |---|---|---|---|---| | -4 | 0.042 | 0.00755 | 5.5 | -0.102 | | -3 | 0.154 | 0.00733 | 20.9 | -0.102 | | -2 | 0.266 | 0.00722 | 36.8 | -0.102 | | -1 | 0.377 | 0.00718 | 52.5 | -0.101 | | 0 | 0.488 | 0.00712 | 68.6 | -0.101 | | 1 | 0.597 | 0.00679 | 87.9 | -0.100 | | 2 | 0.700 | 0.00654 | 107.0 | -0.099 | | 3 | 0.816 | 0.00672 | 121.5 | -0.099 | | 4 | 0.934 | 0.00723 | 129.1 | -0.100 | | 5 | 1.033 | 0.00782 | 132.1 | -0.098 | | 6 | 1.133 | 0.00852 | 133.0 | -0.096 | | 7 | 1.231 | 0.00941 | 130.7 | -0.093 | | 8 | 1.315 | 0.01086 | 121.0 | -0.089 | | 9 | 1.382 | 0.01290 | 107.1 | -0.081 | | 10 | 1.444 | 0.01519 | 95.1 | -0.074 | | 11 | 1.504 | 0.01769 | 85.1 | -0.067 | | 12 | 1.560 | 0.02078 | 75.1 | -0.060 | | 13 | 1.608 | 0.02475 | 64.9 | -0.054 | | 14 | 1.646 | 0.02979 | 55.3 | -0.049 | | 15 | 1.675 | 0.03622 | 46.3 | -0.044 | | 16 | 1.694 | 0.04440 | 38.1 | -0.041 |